Year 9 Mathematics

Unit 4: Constructions

4.3 Constructions 2

Constructions 2

STARTER · 5 MIN

Date: _____ **Title: Constructions 2**

LEARNING OBJECTIVES

- Bisect an angle
- Construct angles of 60°, 30°, 90° and 45°
- Solve problems with constructions

STARTER *Answer in your book. No calculators.*

1. Work out half of 70°
2. Work out half of 90°
3. Each angle in an equilateral triangle?
4. At what angle do a kite's diagonals cross?

Starter — Answers

STARTER

Tick or correct in green pen

1. **35°**
2. **45°**
3. **60°**
4. **90°**

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can bisect an angle	☐
SILVER	I can construct angles of 60°, 30°, 90° and 45°	☐
GOLD	I can solve problems with constructions	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Constructions 2

INTRODUCTION · 3 MIN

An angle bisector cuts an angle exactly in half.

Bisect an angle

Arc from the vertex, then two arcs.

Join the vertex to where they cross.

60°

Use an equilateral triangle.

Same radius from A, then from P.

30° and 45°

Bisect 60° or 90°.

60° ÷ 2 = 30°, 90° ÷ 2 = 45°

Equidistant

Same distance from two lines:

bisect the angle between them.

Examples 1 and 2 — Bisecting Angles

EXAMPLES · 5 MIN

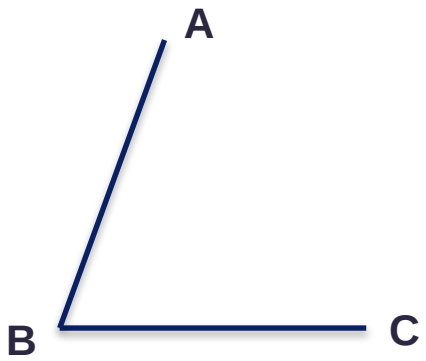
BRONZE

Watch and copy into your book

Hint arc from the vertex, then two equal arcs

Example 1

Construct the bisector of angle ABC.



Example 2

Draw an angle of 110°.

Bisect it. Measure each half.

Examples 1 and 2 — Corrections

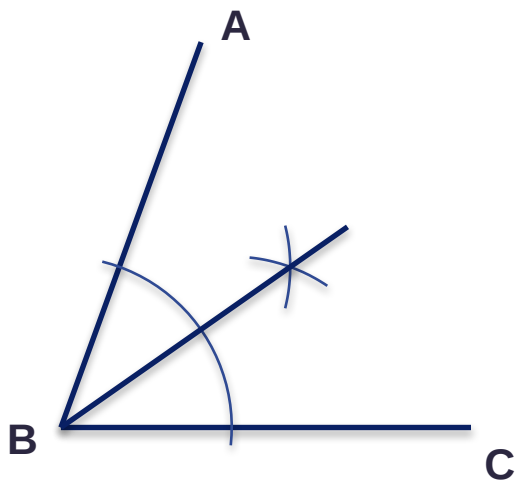
CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

Arc from B, then two equal arcs

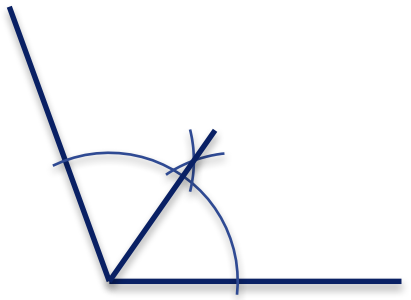


Tip: check with a protractor: 35° each.

Example 2

Same method for any angle

each half = 55°



Tip: keep the compasses the same width.

Task 1

PRACTICE · 9 MIN

BRONZE

Draw each angle with a protractor, then bisect it with compasses

1. 80°
2. 50°
3. 120°
4. 64°
5. 150°

★ 35°

Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

1. **40° each**

2. **25° each**

3. **60° each**

4. **32° each**

5. **75° each**

★ **17.5° each**

Examples 3 and 4 — Constructing Angles

EXAMPLES · 5 MIN

SILVER

Watch and copy into your book

Hint 60°: equilateral arcs 30° = 60° ÷ 2 45° = 90° ÷ 2

Example 3

Construct an angle of 60°. Then bisect it to make 30°.

Example 4

Construct an angle of 90° at P. Then bisect it to make 45°.

Examples 3 and 4 — Corrections

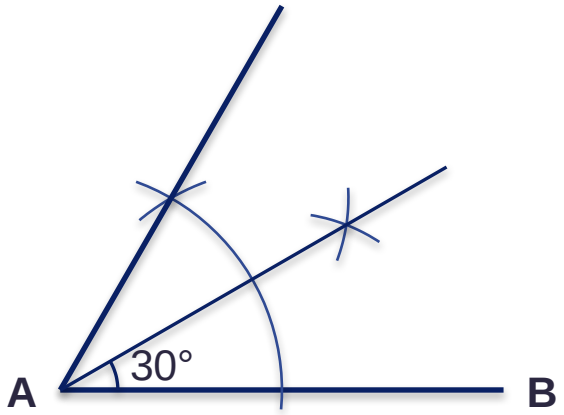
CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

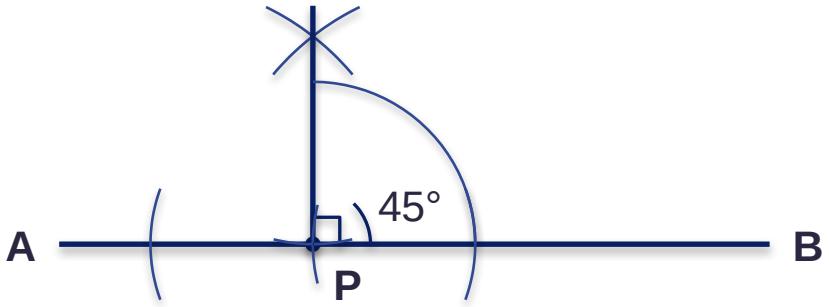
Equilateral arcs, then bisect



Tip: arc from A, then the same arc from P.

Example 4

Perpendicular at P, then bisect



Tip: 90° ÷ 2 = 45°

Task 2

PRACTICE · 9 MIN

SILVER

Construct each angle with a ruler and compasses only. Check with a protractor.

1. 60° at A on a line
2. 30°
3. 90° at P on a line
4. 45°
5. 15°

★ 22.5°

Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1. equilateral arcs
2. bisect 60°
3. perpendicular at P
4. bisect 90°
5. bisect 30°
★ bisect 45°

Examples 5 and 6 — Construction Problems

EXAMPLES · 5 MIN

GOLD

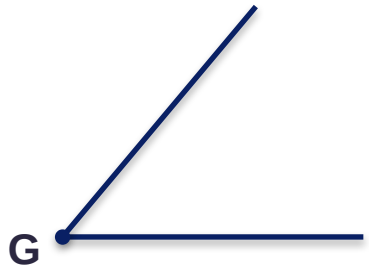
Exam-style: show all your construction arcs

Hint equidistant from two lines: bisect the angle
drawing length = real length ÷ scale

Example 5

(3 marks)

Two rows of trees meet at the gate G. Construct a path from G that is the same distance from both rows.



Example 6

(5 marks)

Kite ABCD: AB = AD = 2.5 m, CB = CD = 4 m, BD = 3 m. Construct it to a scale of 1 cm to 50 cm.

Examples 5 and 6 — Corrections

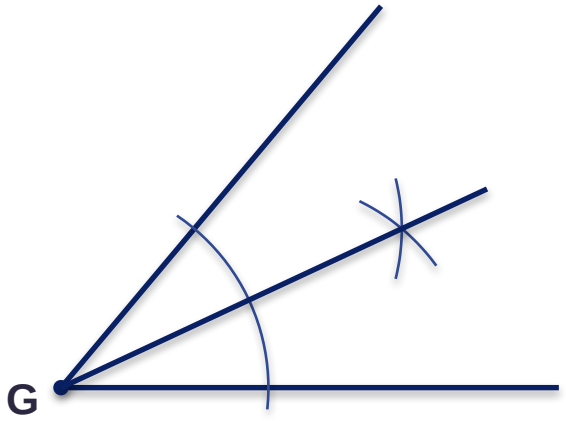
CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

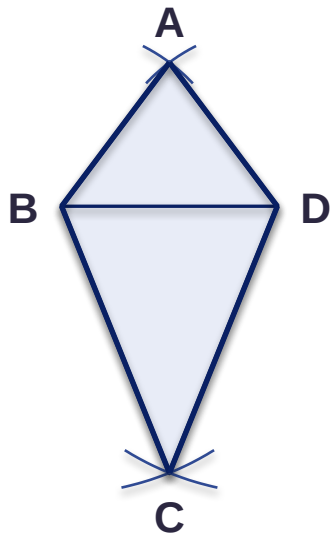
Bisect the angle at G



Tip: every point on the bisector is equidistant.

Example 6

BD = 6 cm, arcs 5 cm and 8 cm



Tip: 2.5 m ÷ 50 cm = 5, so 5 cm.

Task 3

PRACTICE · 9 MIN

GOLD

Construct, showing all arcs. Measure to check.

Hint equidistant from two lines: bisect the angle
drawing length = real length ÷ scale

1. Two walls meet at 80°. A path is the same distance from both walls.

2. Two roads meet at 120°. A fence is the same distance from both roads.

3. Kite: sides 4 cm, 4 cm, 6 cm, 6 cm. The short diagonal is 5 cm.

4. Kite: AB = AD = 3 m, CB = CD = 5 m, BD = 4 m. Scale 1 cm to 1 m.

5. Kite: AB = AD = 1 m, CB = CD = 2 m, BD = 1.5 m. Scale 1 cm to 25 cm.

★ Is the perpendicular bisector of BD a line of symmetry of the kite? Why?

Task 3 — Answers

CORRECTIONS

GOLD *Tick or correct in green pen*

1. **40° with each wall**
2. **60° with each road**
3. **long diagonal ≈ 8.6 cm**
4. **BD 4 cm, AB 3 cm, CB 5 cm**
5. **BD 6 cm, AB 4 cm, CB 8 cm**

★ **Yes: A and C are each the same distance from B and D**

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE Bisect an angle of 100°. What is each half?

SILVER Construct an angle of 30°. Which angle do you bisect?

GOLD A path is equidistant from two fences. What do you draw?

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE **50°**

SILVER **60° (equilateral arcs)**

GOLD **the bisector of the angle between them**

Exam Practice — Bisecting Angles

EXAM PRACTICE · 10 MIN

BRONZE

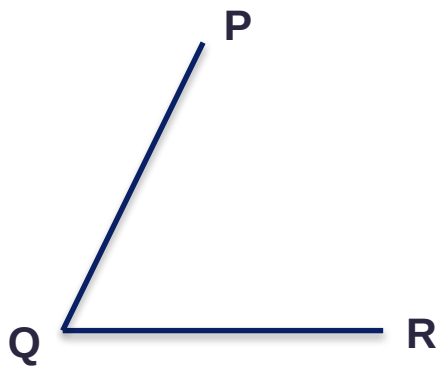
IGCSE exam-style: show all your working

Hint arc from the vertex, then two equal arcs

Question 1

(2 marks)

Construct the bisector of angle PQR.



Question 2

(2 marks)

Draw an angle of 130°.

Construct its bisector.

Measure one of the angles.

..... °

Exam Practice — Constructing Angles

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: show all your working

Hint 60°: equilateral arcs
30° = 60° ÷ 2 45° = 90° ÷ 2

Question 3

(3 marks)

Use a ruler and compasses to construct an angle of 30°.
Show all your construction arcs.

Question 4

(3 marks)

Use a ruler and compasses to construct an angle of 45°.
Show all your construction arcs.

Exam Practice — Construction Problems

EXAM PRACTICE · 10 MIN

GOLD

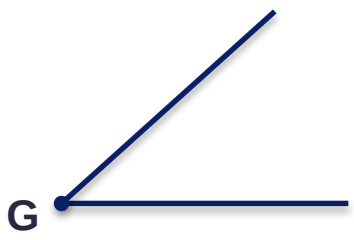
IGCSE exam-style: show all your working

Hint equidistant from two lines: bisect the angle
drawing length = real length ÷ scale

Question 5

(3 marks)

Fergus wants a path from his gate G that is the same distance from the two lines of trees. Construct the path.



Question 6

(6 marks)

Kite ABCD: AB = AD = 2 m,
CB = CD = 3.5 m and BD = 2.5 m.
Ghazi says the perpendicular bisector of BD passes through A and C.

(a) Construct it: 1 cm to 50 cm. (5)

(b) Is Ghazi correct? Why? (1)

.....

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

arc from Q and two equal arcs — M1

bisector drawn — A1

Tip: each half should be 32°.

Question 2

correct arcs — M1

65° — A1

Tip: 130° ÷ 2 = 65°

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

60° constructed with equal arcs	M1
arcs to bisect it	M1
30° drawn	A1

Tip: 30° is half of 60°.

Question 4

90° constructed	M1
arcs to bisect it	M1
45° drawn	A1

Tip: 45° is half of 90°.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

arc from G cutting both lines	M1
two equal arcs	M1
bisector drawn from G	A1

Tip: the path bisects the angle at G.

Question 6

(a) BD = 5 cm drawn	M1
arcs 4 cm and 7 cm from B, D	M3
kite completed	A1
(b) Yes: A and C are each	
equidistant from B and D	B1

Tip: divide each length by 50 cm.